

Ibn Tofail University

Rigid Body Mechanics

Problem Set III

Note: All solids are assumed to be homogeneous and of total mass m .

Exercise 1:

Determine the center of mass (center of inertia) of the following solids:

- 1) A homogeneous circular arc of radius R and angle 2α .
- 2) A homogeneous circular plate of radius R and angle 2α .
- 3) Hemisphere of center O and radius R .
- 4) Solid cone.
- 5) Solid cone surmounted by a hemisphere.

Correction

Exercise 2:

Determine the inertia matrix at O relative to the reference frame $R(O, X, Y, Z)$ of a quarter disk of radius R . The axes $\overrightarrow{OX}$ and $\overrightarrow{OY}$ are on the edges of the quarter disk. Calculate the moment of inertia with respect to the line (Δ) passing through point O and point A with coordinates $(R, R, 0)$.

Correction

Exercise 3:

Determine the inertia matrix of the following solids:

- 1) A rod (OA) of length L (at extremity O " I_O " and at center of mass G " I_G ").
- 2) A rectangular plate (OABC) with sides a and b (at center of mass G and at point O).
- 3) Solid rectangular parallelepiped with dimensions a , b , c and center of mass G (at center G).
- 4) A solid cylinder (C), of radius R and height h (at the center of its base O and at center of mass G).
- 5) Disk (D) with center of inertia G and radius R .
- 6) Hemisphere (B) of radius R (at the center of its base O) " I_O ".
- 7) A sphere (B) with center of inertia G and radius R (at center of mass G).

Correction